Electric Vehicle Adoption and Market Trend Analysis Using Data Science in R

GUNDLAPALLY ANANYA (011)

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1. Introduction

Electric vehicles (EVs) are becoming an important part of the transition toward sustainable transportation. This case study uses R to analyze EV sales trends, compare EV sales shares across countries, examine charging infrastructure, and forecast future global EV sales.

2. Problem Statement

EV adoption varies across countries and over time. Analyzing historical sales, country-level adoption, and charging infrastructure can help identify trends and estimate future EV sales.

3. Objectives

- Analyze historical global EV sales trends.
- Compare EV sales shares across countries.
- Examine global EV charging infrastructure.
- Forecast global EV sales and evaluate forecasting models.

4. Dataset Description

The dataset contains EV statistics by country or region, category, parameter, mode, powertrain, year, unit, and value. It includes historical data and projections for selected indicators.

5. Data Loading and Preprocessing

```
file_path <- "C:/Users/anany/Downloads/EV data by country 2026.xlsx"

ev_data <- read_excel(
  file_path,
  sheet = "GEVO_EV_2026"
)

# Inspect the original dataset
dim(ev_data)
```

```
## [1] 49375     9
```

```
str(ev_data)
```

```
## tibble [49,375 × 9] (S3: tbl_df/tbl/data.frame)
##  $ region_country : chr [1:49375] "Chile" "Norway" "Portugal" "Canada" ...
##  $ category       : chr [1:49375] "Historical" "Historical" "Historical" "Historical" ...
##  $ parameter      : chr [1:49375] "EV charging points" "EV charging points" "EV charging p
oints" "EV charging points" ...
##  $ mode           : chr [1:49375] "EVSE" "EVSE" "EVSE" "EVSE" ...
##  $ powertrain     : chr [1:49375] "Publicly available fast" "Publicly available fast" "Pub
licly available fast" "Publicly available fast" ...
##  $ year           : num [1:49375] 2011 2011 2011 2012 2012 ...
##  $ unit           : chr [1:49375] "charging points" "charging points" "charging points" "c
harging points" ...
##  $ value          : num [1:49375] 1 18 6 2 4 9 18 1 21 58 ...
##  $ Aggregate group: chr [1:49375] "Other" "Other" "Other" "Other" ...
```

```
summary(ev_data)
```

```
##    region_country       category         parameter            mode
##  Length   :49375   Length   :49375   Length   :49375   Length   :49375
##  N.unique :   72   N.unique :    3   N.unique :   19   N.unique :   13
##  N.blank  :    0   N.blank  :    0   N.blank  :    0   N.blank  :    0
##  Min.nchar:    3   Min.nchar:   10   Min.nchar:    8   Min.nchar:    2
##  Max.nchar:   32   Max.nchar:   16   Max.nchar:   32   Max.nchar:   17
##  NAs      : 1078
##      powertrain         year             unit           value
##  Length   :49375   Min.   :2010   Length   :49375   Min.   :        0
##  N.unique :   10   1st Qu.:2015   N.unique :    9   1st Qu.:       60
##  N.blank  :    0   Median :2019   N.blank  :    0   Median :    17000
##  Min.nchar:    2   Mean   :2019   Min.nchar:    3   Mean   :   320318
##  Max.nchar:   29   3rd Qu.:2022   Max.nchar:   34   3rd Qu.:    62000
##                    Max.   :2035   NAs      :22582   Max.   :470000000
##   Aggregate group
##  Length   :49375
##  N.unique :    6
##  N.blank  :    0
##  Min.nchar:    5
##  Max.nchar:   18
##
```

```
# Check missing values
colSums(is.na(ev_data))
```

```
##  region_country        category       parameter            mode      powertrain
##            1078               0               0               0               0
##            year            unit           value Aggregate group
##               0           22582               0               0
```

```
# Remove duplicate rows
ev_data <- ev_data %>%
  distinct()

# Verify the cleaned dataset
dim(ev_data)
```

```
## [1] 49360     9
```

The original dataset contained 49,375 records and 9 columns. The inspection identified 1,078 missing values in `region_country` and 22,582 missing values in `unit`. No missing values were found in the other columns. After removing duplicate rows using `distinct()`, the dataset contained 49,360 records.

Missing values were retained during this initial cleaning stage because their treatment depends on the analysis being performed. Records with missing country names or units should be handled appropriately when filtering and interpreting individual analyses.

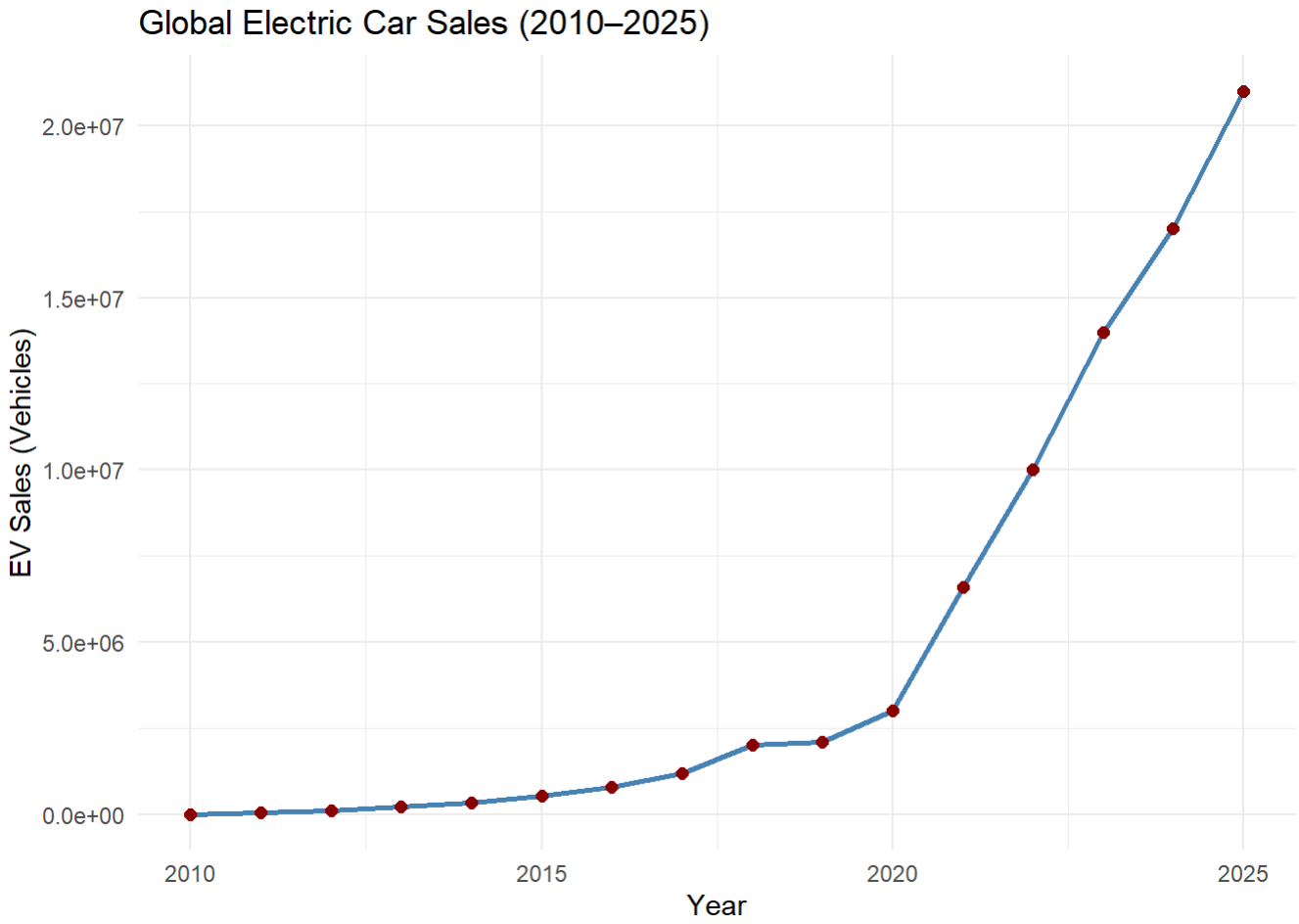
6. Global EV Sales Trend Analysis

```
global_sales <- ev_data %>%
  filter(
    region_country == "World",
    category == "Historical",
    parameter == "EV sales",
    mode == "Cars",
    powertrain == "EV"
  ) %>%
  select(year, value) %>%
  distinct(year, .keep_all = TRUE) %>%
  arrange(year)

print(global_sales)
```

```
## # A tibble: 16 × 2
##     year    value
##    <dbl>    <dbl>
##  1  2010     7400
##  2  2011    49000
##  3  2012   120000
##  4  2013   210000
##  5  2014   330000
##  6  2015   530000
##  7  2016   780000
##  8  2017  1200000
##  9  2018  2000000
## 10  2019  2100000
## 11  2020  3000000
## 12  2021  6600000
## 13  2022 10000000
## 14  2023 14000000
## 15  2024 17000000
## 16  2025 21000000
```

```
ggplot(global_sales, aes(x = year, y = value)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point(color = "darkred", size = 2) +
  labs(
    title = "Global Electric Car Sales (2010–2025)",
    x = "Year",
    y = "EV Sales (Vehicles)"
  ) +
  theme_minimal()
```



Interpretation

The analysis examines historical global electric car sales from 2010 to 2025. The trend indicates substantial growth in EV sales over the period, with a particularly strong increase in recent years. This highlights the expanding adoption of electric cars globally.

7. Country-wise EV Adoption Analysis

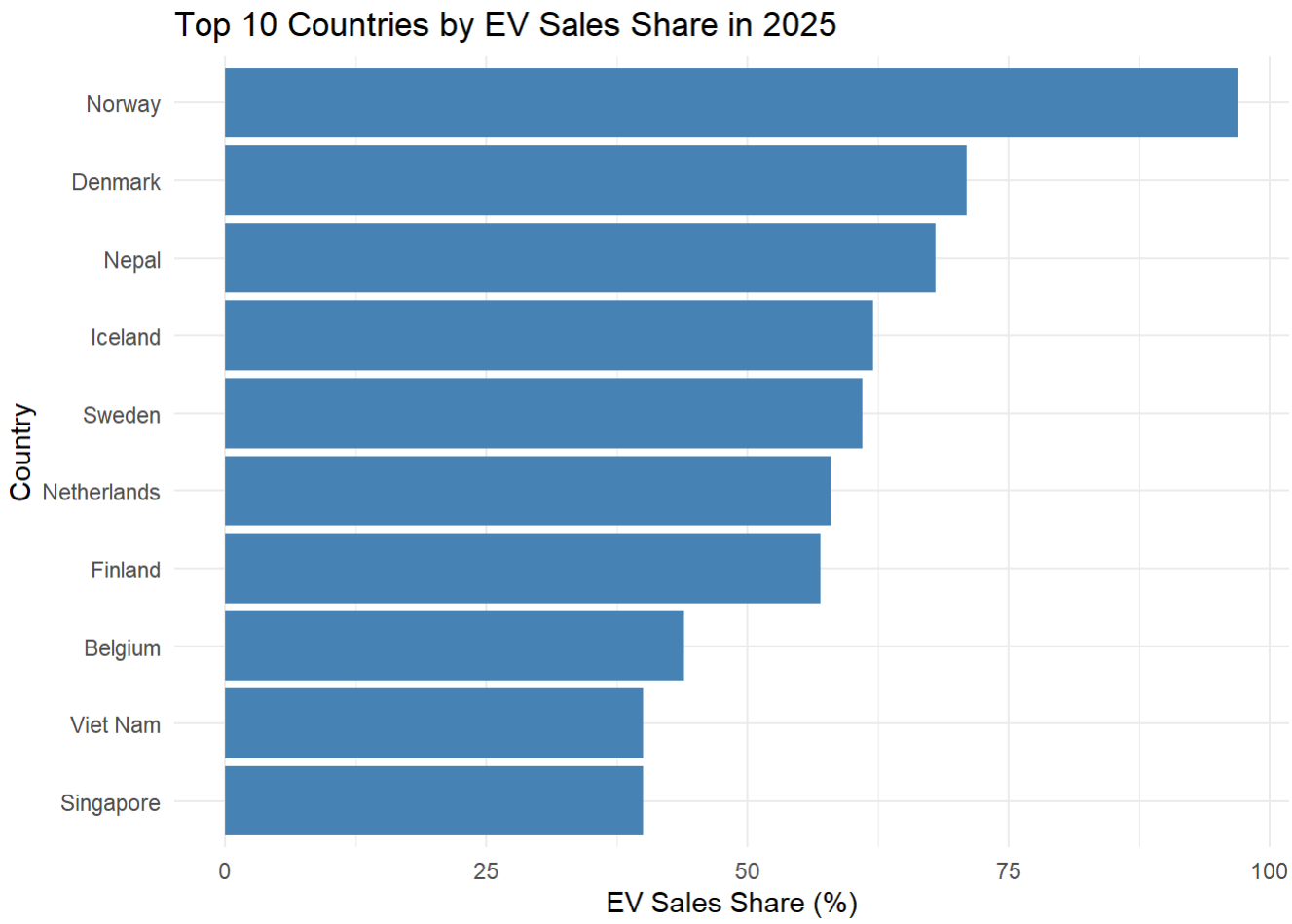
```r
country_ev_clean <- ev_data %>%
  filter(
    category == "Historical",
    parameter == "EV sales share",
    mode == "Cars",
    powertrain == "EV",
    year == 2025,
    `Aggregate group` == "Other",
    !is.na(region_country)
  ) %>%
  select(region_country, value) %>%
  distinct() %>%
  arrange(desc(value))

top_countries_clean <- country_ev_clean %>%
  slice_head(n = 10)

print(top_countries_clean)
```

```
## # A tibble: 10 × 2
##    region_country value
##    <chr>          <dbl>
##  1 Norway            97
##  2 Denmark           71
##  3 Nepal             68
##  4 Iceland           62
##  5 Sweden            61
##  6 Netherlands       58
##  7 Finland           57
##  8 Belgium           44
##  9 Singapore         40
## 10 Viet Nam          40
```

```r
ggplot(
  top_countries_clean,
  aes(x = reorder(region_country, value), y = value)
) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Top 10 Countries by EV Sales Share in 2025",
    x = "Country",
    y = "EV Sales Share (%)"
  ) +
  theme_minimal()
```



Interpretation

The country-wise comparison identifies differences in EV sales share among the ten countries displayed. Norway has the highest share in this analysis, followed by Denmark and Nepal. EV sales share measures the proportion of new car sales that are electric.

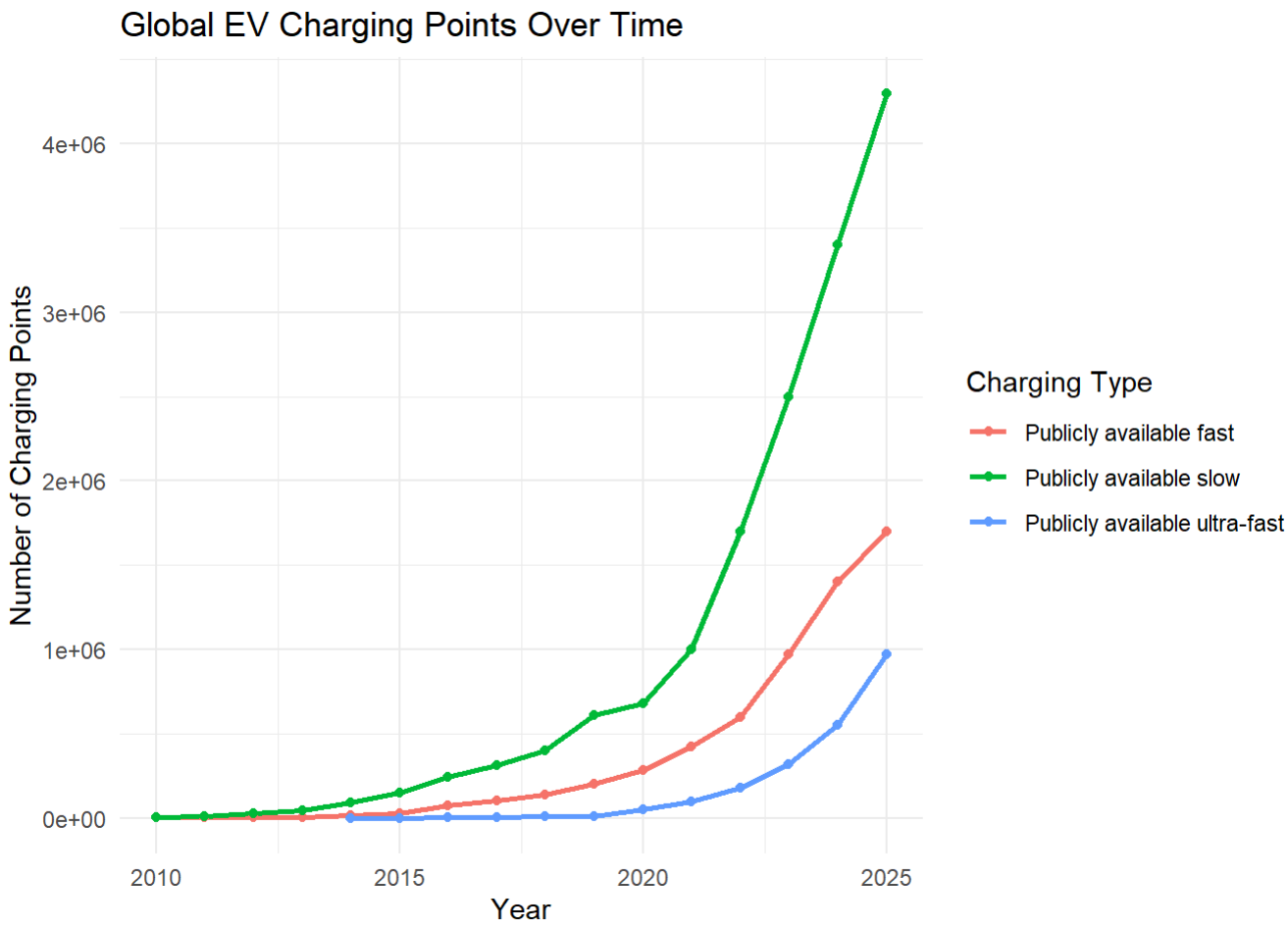
8. EV Charging Infrastructure Analysis

```
charging_data <- ev_data %>%
  filter(
    region_country == "World",
    category == "Historical",
    parameter == "EV charging points"
  ) %>%
  select(year, powertrain, value) %>%
  distinct() %>%
  arrange(year)

print(head(charging_data, 10))
```

```
## # A tibble: 10 × 3
##     year powertrain                value
##    <dbl> <chr>                     <dbl>
##  1  2010 Publicly available fast   310
##  2  2010 Publicly available slow  3700
##  3  2011 Publicly available fast   830
##  4  2011 Publicly available slow 10000
##  5  2012 Publicly available fast  1600
##  6  2012 Publicly available slow 29000
##  7  2013 Publicly available fast  2600
##  8  2013 Publicly available slow 42000
##  9  2014 Publicly available fast 13000
## 10  2014 Publicly available slow 90000
```

```
ggplot(
  charging_data,
  aes(x = year, y = value, color = powertrain)
) +
  geom_line(linewidth = 1) +
  geom_point(size = 1.5) +
  labs(
    title = "Global EV Charging Points Over Time",
    x = "Year",
    y = "Number of Charging Points",
    color = "Charging Type"
  ) +
  theme_minimal()
```



Interpretation

This analysis examines historical changes in global EV charging infrastructure. Comparing charging categories helps identify how the availability of different types of charging points has changed over time. Charging infrastructure is important for supporting the adoption and use of electric vehicles.

9. EV Sales Forecasting and Model Evaluation

9.1 Prepare training and testing data

```
sales_data <- ev_data %>%
  filter(
    region_country == "World",
    category == "Historical",
    parameter == "EV sales",
    mode == "Cars",
    powertrain == "EV"
  ) %>%
  select(year, value) %>%
  distinct(year, .keep_all = TRUE) %>%
  arrange(year)

train_data <- sales_data %>%
  filter(year <= 2022)

test_data <- sales_data %>%
  filter(year >= 2023, year <= 2025)

train_ts <- ts(
  train_data$value,
  start = min(train_data$year),
  frequency = 1
)
```

9.2 Fit the forecasting models

```
# Baseline: use the last training observation
baseline_pred <- rep(
  tail(train_data$value, 1),
  nrow(test_data)
)

# ARIMA model
arima_model <- forecast::auto.arima(
  log(train_ts),
  seasonal = FALSE
)

arima_pred <- exp(as.numeric(
  forecast::forecast(arima_model, h = nrow(test_data))$mean
))

# ETS model
ets_model <- forecast::ets(log(train_ts))

ets_pred <- exp(as.numeric(
  forecast::forecast(ets_model, h = nrow(test_data))$mean
))

comparison <- data.frame(
  Year = test_data$year,
  Actual = test_data$value,
  Baseline = baseline_pred,
  ARIMA = arima_pred,
  ETS = ets_pred
)

print(comparison)
```

```
##   Year  Actual Baseline    ARIMA      ETS
## 1 2023 1.4e+07    1e+07 18234647 13743305
## 2 2024 1.7e+07    1e+07 33250235 17724145
## 3 2025 2.1e+07    1e+07 60630628 21724241
```

9.3 Evaluate model performance

```
evaluate_model <- function(actual, predicted) {
  data.frame(
    MAE = mean(abs(actual - predicted)),
    RMSE = sqrt(mean((actual - predicted)^2)),
    MAPE = mean(abs((actual - predicted) / actual)) * 100
  )
}

baseline_metrics <- evaluate_model(
  comparison$Actual, comparison$Baseline
)

arima_metrics <- evaluate_model(
  comparison$Actual, comparison$ARIMA
)

ets_metrics <- evaluate_model(
  comparison$Actual, comparison$ETS
)

model_comparison <- data.frame(
  Model = c("Baseline", "ARIMA", "ETS"),
  MAE = c(
    baseline_metrics$MAE,
    arima_metrics$MAE,
    ets_metrics$MAE
  ),
  RMSE = c(
    baseline_metrics$RMSE,
    arima_metrics$RMSE,
    ets_metrics$RMSE
  ),
  MAPE = c(
    baseline_metrics$MAPE,
    arima_metrics$MAPE,
    ets_metrics$MAPE
  )
)

print(model_comparison)
```

```
##      Model        MAE       RMSE      MAPE
## 1 Baseline  7333333.3  7874007.9  40.70962
## 2    ARIMA 20038503.3 24850144.7 104.85146
## 3      ETS   568360.4   609590.9   3.18066
```

10. Global EV Sales Forecast (2026–2030)

```r
# Fit the ETS model using all historical sales data
sales_ts <- ts(
  sales_data$value,
  start = min(sales_data$year),
  frequency = 1
)

ets_final_model <- forecast::ets(log(sales_ts))

# Forecast the next five years
ets_final_forecast <- forecast::forecast(
  ets_final_model,
  h = 5,
  level = 95
)

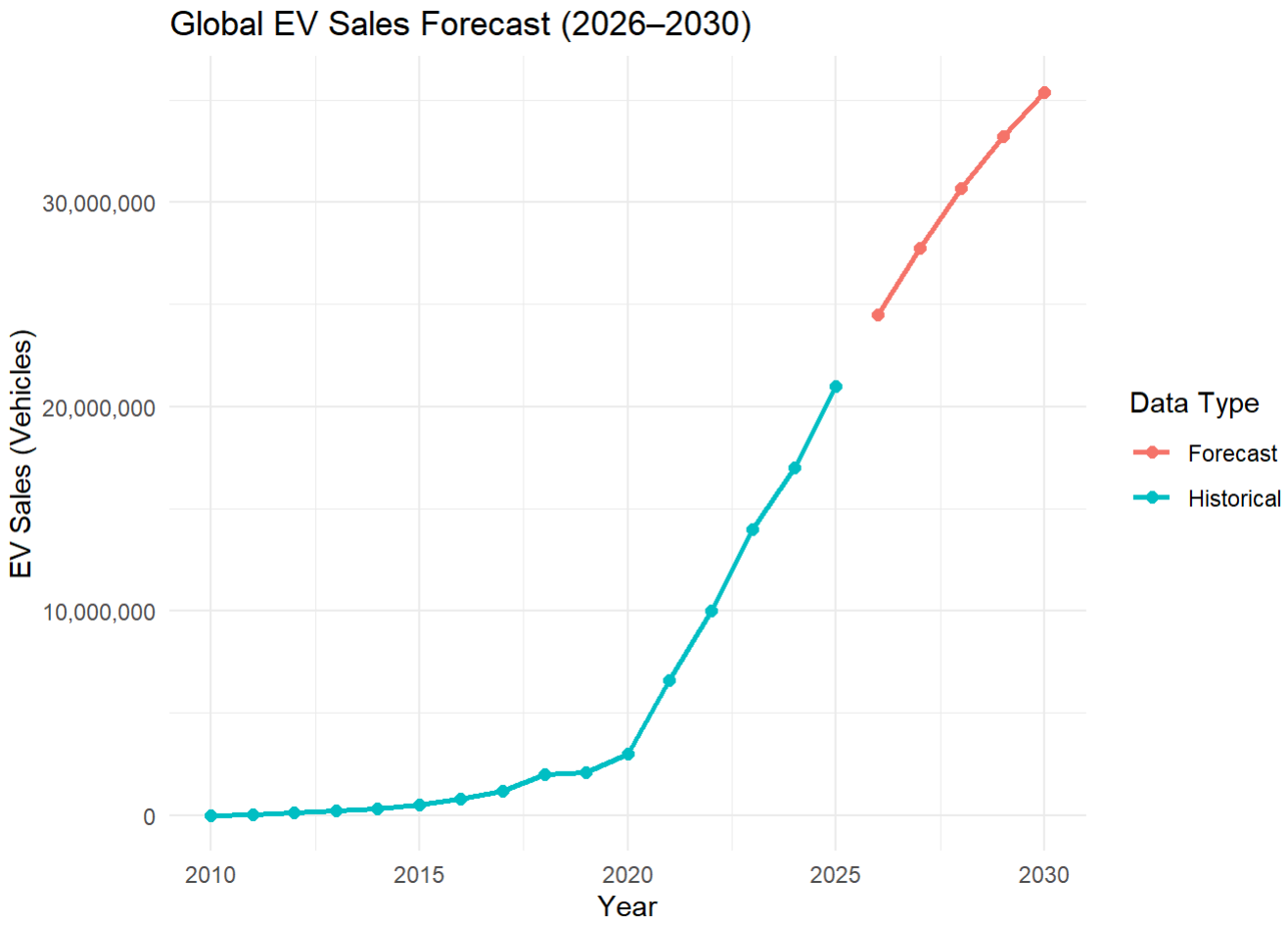
# Convert predictions and intervals to vehicle counts
forecast_values <- data.frame(
  Year = 2026:2030,
  Predicted_Sales = round(exp(as.numeric(ets_final_forecast$mean))),
  Lower_95 = round(exp(as.numeric(ets_final_forecast$lower[, 1]))),
  Upper_95 = round(exp(as.numeric(ets_final_forecast$upper[, 1])))
)

print(forecast_values)
```

```
##   Year Predicted_Sales Lower_95  Upper_95
## 1 2026        24511909 13324987  45090751
## 2 2027        27761391 10352398  74446020
## 3 2028        30668497  7913427 118855800
## 4 2029        33211883  5980150 184448410
## 5 2030        35397640  4494894 278759180
```

```r
# Prepare historical and forecast values for plotting
forecast_plot_data <- data.frame(
  Year = c(sales_data$year, 2026:2030),
  Sales = c(
    sales_data$value,
    round(exp(as.numeric(ets_final_forecast$mean)))
  ),
  Type = c(
    rep("Historical", nrow(sales_data)),
    rep("Forecast", 5)
  )
)

# Plot historical sales and forecasts
ggplot(
  forecast_plot_data,
  aes(x = Year, y = Sales, color = Type)
) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    title = "Global EV Sales Forecast (2026–2030)",
    x = "Year",
    y = "EV Sales (Vehicles)",
    color = "Data Type"
  ) +
  scale_y_continuous(labels = scales::label_comma()) +
  theme_minimal()
```



Interpretation

The ETS model estimates future global electric car sales using historical data from 2010 to 2025. The forecast table presents predicted sales for 2026–2030 and their 95% prediction intervals. The intervals indicate uncertainty, which generally increases over the forecast horizon. These estimates are model-based projections rather than guaranteed outcomes.

11. Conclusion

This case study analyzed global electric car sales, country-wise EV sales shares, and charging infrastructure using R. The results showed substantial growth in global EV sales between 2010 and 2025, along with differences in EV adoption across countries. The charging infrastructure analysis also demonstrated changes in publicly available fast and slow charging points.

Among the Baseline, ARIMA, and ETS forecasting models evaluated on the 2023–2025 test period, ETS achieved the lowest MAE, RMSE, and MAPE. The final ETS model was used to forecast global EV sales through 2030. However, the short test period and increasing forecast uncertainty limit the reliability of long-term predictions.

12. Limitations and Future Scope

Limitations

- The forecasting test set contains only three years.
- Predictions based on historical trends may not capture future policy, economic, technological, or market changes.
- Missing country names and units may affect some analyses.
- Forecasts are estimates and should not be interpreted as guaranteed outcomes.

Future Scope

- Evaluate forecasting models using longer validation periods.
- Include additional variables such as vehicle prices, charging availability, government policies, and economic indicators.
- Compare additional forecasting approaches and validate them on unseen data.

13. References

1. International Energy Agency (IEA). (2026). *Global EV Outlook 2026: EV data by country*. https://www.iea.org/data-and-statistics/data-product/global-ev-outlook-2026 (https://www.iea.org/data-and-statistics/data-product/global-ev-outlook-2026)
2. R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/ (https://www.R-project.org/)
3. RStudio. *R Markdown Documentation*. https://rmarkdown.rstudio.com/ (https://rmarkdown.rstudio.com/)
4. Hyndman, R. J., Athanasopoulos, G., and others. *forecast: Forecasting Functions for Time Series and Linear Models*. R package documentation. https://pkg.robjhyndman.com/forecast/ (https://pkg.robjhyndman.com/forecast/)